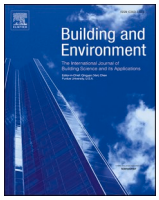
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Applicability of a novel Indicator Method to assessment of the impact of buildings on the environment

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ABSTRACT

It is common in the building sector to require that several variants of a planned building be developed. Evaluation of such variant solutions can refer to different aspects, but in compliance with the binding EU legal regulations, it is obligatory to take into consideration different stages in the life cycle of a building in any such assessment. As recommended, the evaluation of a life cycle should include three basic aspects, that is costs and the impact on the social and natural environment. It can encompass the development of an entire building, or in other cases it might be justified to analyse particular components of a development project.

The literature on the subject implicates certain shortcomings of commonly used methods. The article presents an innovative approach to the assessment of variants, based on a case of designing a roof of a production hall, where various material solutions for the roof truss are considered. A proprietary approach called the Indicator Method was used for the evaluation. The method employs both positive and negative values, direct and indirect (local and global) impacts, and the model appraisal was based on selected indicators for all stages of the facility's life cycle. Assessment with negative and positive values yields more varied results, which may facilitate decision making. Furthermore, the inclusion of factors covering all stages of the life cycle has identified critical stages in the facility's life cycle. The research confirmed that the proposed Indicator Method can fill identified the research gap by providing innovative tools to evaluate the environmental impact of a building more comprehensively.

1. Introduction

Preparation of construction projects most often requires the development and evaluation of several variants of their implementation [1–3], including many materials, technologies and construction solutions [4–6]. On the one hand, this proceeding is enforced by the binding regulations, e.g. on public procurement or using EU funds [7–9]. On the other hand, every investor who allocates their funds to a construction development would like to do it in the best possible way, so that the future building will guarantee long and trouble-free use. Different evaluation criteria can be involved in making an assessment of the prepared variants. The criteria actually used are selected according to the expectations and requirements that a given building should satisfy [10–12]. Obviously, the requirements depend on many factors, for example the function that the constructed building will play and its use, conditions in which it will be used (both external and internal), and the time period for which its use is planned [13,14]. Out of these factors, a group of criteria emerges that can be employed for making an assessment. Many criteria are typically used, and the evaluation of construction variants can be supported by multi-criteria methods. The literature [15–17] describes many methods applied for an analysis of variants, but only a few, simple ones, based on scores or weights and scores, are used in practice. Evaluations of variants should be carried out in such a way as to take into account all stages in the life cycle of buildings. However, when making an assessment of variants, the most popular criteria applied are divided into groups like economic, technical, functional and environmental criteria [18,19]. Details of an analysis and a set of employed factors depend on the type of a building, conditions in which it will be constructed, and its later use. It is difficult to find examples of analyses of design variants presenting groups of criteria assigned to particular stages in a building's life cycle, although this approach to the problem could be useful in analyses, leading to the selection of the most advantageous variant for the investor, developer or user. It could also help to indicate the time period when special attention should be paid in order to optimise the processes which should be executed during the given stage [20–22]. An assessment of variants based on groups of criteria assigned to subsequent stages in the life cycle could be particularly helpful in evaluating the impact of a planned

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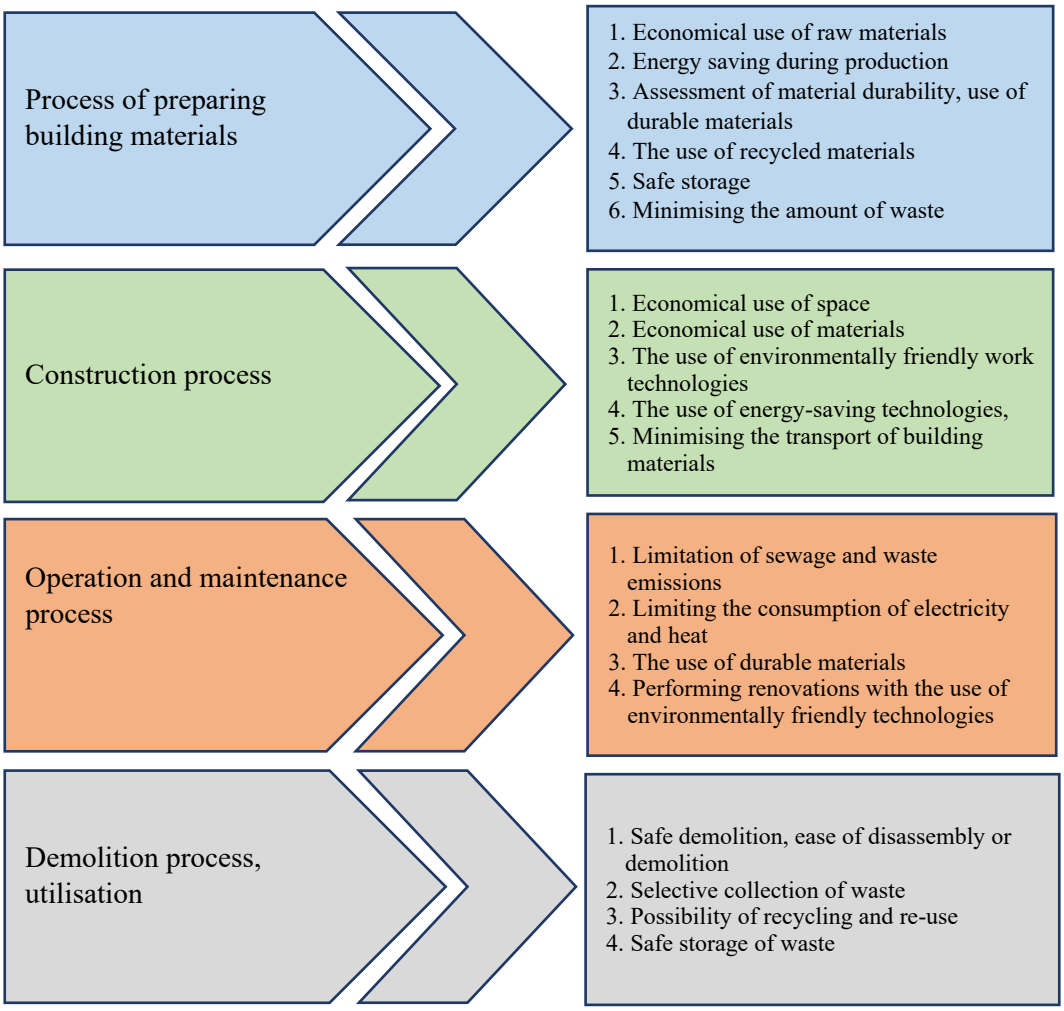


Fig. 1. Examples of criteria for assessing environmental impacts, taking into account the stages of the facility's life cycle.

development on the natural environment. The literature contains many reports on analyses including the life cycle of specific materials and construction industry products, while fewer articles are dedicated to the life cycle of entire buildings [23–25].

Nowadays, when environmental protection has become a priority, new concepts of pro-ecological behaviour are emerging [26–28]. One of the proposed solutions is green technology [29–31]. In construction, green technology entails primarily an effective management of energy and natural resources, from production of building materials and energy-saving technologies, through design solutions and technologies used during the construction works, to the useable life of a building. Another important issue is how to dispose of and recycle demolition waste. All such activities should be viewed from the perspective of a building's life cycle [32–34].

A significant role of environmental protection can also be noticed in evaluations of variant designs of buildings. Criteria classified as environmental ones are gaining importance, and weights allocated to them are often higher than the ones attributed to economic and technical criteria. However, it is important to assign all factors describing the environmental effects to specific stages in the life cycle of a building.

It is noticeable in relevant literature [22,24,26] that the evaluation of the building-environment relationship usually concerns one selected period, and most often it is the operation stage. This is understandable because it is the longest phase, but omitting the other life-cycle stages from an assessment may yield an erroneous outcome. It is also common practice in impact assessments to assign importance to criteria on a predetermined scale, usually with scores of 1–5, and sometimes 1–2 or 1–7. This method is often extended so as to evaluate the extent to which specific criteria (or impacts) are met by the appraised options. Such a procedure often gives poorly differentiated results [17,18]. In the studies conducted by the authors, negative values were added to environmental impact assessments. The use of an extended evaluation scale, e.g. from −5 to +5, can generate more diverse results and indicate solutions with a positive or negative impact on the environment. None of the approaches previously described in the literature has employed a scale with negative scores.

2. Impacts of a construction development in a building's life cycle

An analysis covering the whole life cycle of a building generates several factors, which can be submitted to further evaluations [35,36]. They can be described and classified in line with the division into basic life cycle stages, including an option to distinguish certain processes, like the process of preparing building materials, the construction works, the process of using a building, and the process of liquidating a disused building [37–39]. The observations made during the research substantiated the specification of a group of criteria with an impact on the final outcome of an assessment. Fig. 1 provides a list of the most important criteria which feature in LCA analyses [40–42].

2.1. Ecological evaluation of building materials and the process of preparing

Because the construction sector is this branch of economy that most probably consumes most raw and semi-finished materials globally, it is essential to draw attention to the proper management of materials, optimisation of the production of building materials, and consumption of energy due to the production and transport of building materials [37,38]. The ecological evaluation of this stage comprises the following factors:

- economical use of natural resources – protection of fossil raw materials, use of renewable resources, use of recycled materials;
- elimination of hazardous chemical and biological contaminations – minimising the emission of toxins to the environment, avoiding the contamination dangerous to people in cases of malfunctions; absence of any threats during the production of a building material and after it has been used for construction works;
- optimisation of energy consumption during the process of making, transporting and using building materials;
- saving energy for heating a building by using materials which will lower the building's demand for energy;
- evaluation of the durability of a building material – long service of the material without compromising its characteristics, minimising the need for future repairs;
- recovery of the material and its possible recycling – easy disassembly or demolition, possibility of re-using the material;
- minimising quantities of waste under normal conditions, without the need to set up special disposal sites, safe disposal and storage of waste.

As noticeable, several factors defined in connection with the production of building materials link their evaluation to the future time of using a building as well as its later demolition and recycling of used materials [36,38].

2.2. Evaluation of the process of constructing a building

The process of constructing a building is the shortest period in the building's life cycle [37,38,40]. Its evaluation is comprised in a few points:

- economical use of the area and protection of natural areas adjacent to the building site;
- economical use of building materials;
- construction works carried out by taking into account the processes friendly to the natural environment;
- use of energy-saving technologies;
- minimising the transport of building materials at a construction site.

2.3. Evaluation of the process of using a building

The process of using a building is the longest stage in the building's life cycle, encompassing many aspects. Above all, it is associated with two groups of activities: operations connected with the ongoing use of a building, and maintenance and repairs to keep the building in a useable condition [37,40]. The following factors can be listed:

- reducing the emission of wastewater and waste as well as consumption of energy;
- using durable materials, reducing maintenance works and repairs;
- making repairs using eco-friendly technologies;
- the longest possible use of buildings, taking care of existing buildings.

2.4. Evaluation of the process of demolition, possibilities of recycling and disposal of demolition materials

Evaluation of the possibilities of recycling demolition materials and possible problems with their recovery are often the most important points in LCA evaluations [38,40,42]. This issue has attracted much attention in the EU regulations and directives. The most important guidelines lead to the selection of the following points in an assessment:

- demolition works can be carried out safely;
- demolition materials are collected selectively and therefore can be re-used;
- recovery of building materials and their recyclability are possible – easy disassembly or demolition;
- safe storage of the waste can be ensured, minimising amounts of waste, thus making disposal sites unnecessary.

The factors presented above demonstrate the immense diversity of problems related to the impact on the environment throughout the whole life cycle of a building. Thus, only some criteria, significant in a given case, are chosen for an assessment. Noteworthy is the fact that the factors considered at the stage of producing building materials in most cases refer to the subsequent stages of an assessment. For example, the economical use of raw materials, lack of negative biological or chemical impacts or recyclability arise from the assumptions made at the stage of manufacturing materials. Buildings can serve highly different functions, have a long life span, and heavily interfere with the surroundings. A debatable question regarding buildings under construction is how to determine the condition of any such building once its useable life terminates. As for old buildings, it is difficult to obtain accurate information about the stage of their construction. To make an analysis, it is necessary to delineate boundaries of the analysed system, which are closely dependent on its functions.

All life cycle stages as well as all circumstances of preparing a construction development, its implementation and then use of a ready building can serve as a basis for defining the criteria for an evaluation of the variants of the building's design. Due to the complex nature of the problem, there is truly a large number of such criteria. Hence, such evaluations employ multi-criteria methods of analysis. However, not all methods presented in the literature are useful for every type of an appraisal, and this is true for the kind of an assessment discussed in this paper. A review of such methods is presented in the following section of this article, which ends with the suggestion which method satisfies the best the expectations set for the assessment in question.

3. Review of multi-criteria methods

The theory of the evaluation of an engineering solution based on many criteria, following many years of its development, has led to the emergence of numerous and highly varied methods. At present, there is a notable diversity and distinct discrepancies among the names of the methods and their classification [43,44]. The multi-criteria decision support approach, multi-attribute decision theory, multi-criteria analysis are just a few of the terms used by researchers and experts [45–47]. Several methods, which have been designed after years of research conducted by scholars engaged in multi-criteria analyses [45–47], can now be grouped as:

- scalarisation methods,
- outranking methods with preference relations,
- distance function methods,
- lexicographic and hierarchical methods,
- group analysis methods.

Another classification, with guidelines for the terminology in multi-criteria analyses, is proposed by the authors [48,49]. Based on their

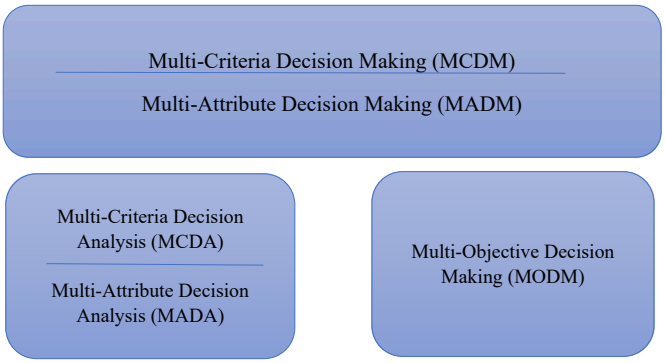


Fig. 2. Classification of multi-criteria methods (Source: the authors, based on [49]).

perusal of source materials, they placed the group of multi-criteria methods under the umbrella of Multi-Criteria Decision Making (MCDM) methods, acknowledging also the less common name: Multi-Attribute Decision Making (MADM). The MCDM/MADM approaches can be divided into two groups, viewing a decision-making problem in two different ways [49,50]. One group comprises Multi-Criteria Decision Analysis (MCDA), and Multi-Attribute Decision Analysis (MADA), where an assessment involves established, specific and previously described variants. The other group is Multi-Objective Decision Making (MODM), where the methods enable the user to generate variants by describing them with the values assigned to appropriate parameters. The most significant difference between the MCDA and MODM methods is the character of the domain of variants and criteria. In MCDA, a set of possible solution is decreet and the criteria do not have to be quantifiable. The other approach represented by the MODM group of methods is where the domain of variants is a continuous set, analysed with the help of calculable features and parameters. Investment decision support in the building sector usually consists in an evaluation of a discreet group of variants. The classification of multi-criteria methods [49] is presented schematically in Fig. 2.

3.1. Methods of multi-criteria analysis most often employed in the building sector

With a wide array of methods and approaches to multi-criteria analyses, it is possible to choose a procedure that will best fit specific decision-making situations. Building investments and construction projects have their own specific character, which excludes some methods from being used in practice. The most important feature that a decision-support methods employed in the building practice should possess is to enable an appraisal of a large number of criteria of various nature [48,49]. At the stage of comparing variants, there is a need determine the degree to which the required parameter is satisfied rather than make a binary assessment. A significant requirement is that descriptive assessment scales need to be developed clearly enough to facilitate their application. Of importance in applying any of these methods is also having a simple data curation instruction, uncomplicated mathematical apparatus, and understandable presentation of the result. Therefore, there is a certain group of methods which seem more popular than others in the building sector [50–52]. Analyses of investments into building developments where alternative solutions have been evaluated demonstrate that the methods most often used are the ones where assessments are made by assigning scores. Fig. 3 collates the statistical data showing the popularity of methods employed for evaluation of development projects financed from the EU funds in north-eastern Poland in 2015–2020. Among the analysed investments, carried out in this part of Poland, the simplest decision-support methods, i.e. score and weight-score evaluations, prevailed owing to their simplicity. The AHP method is among the most mathematically advanced approaches used in practice [53,54].

3.2. Research problem

The multi-criteria analysis methods used to evaluate variants of construction projects are universal methods. What the mentioned methods share is the use of a score scale for making an assessment to what extent particular criteria are satisfied by the assessed solution. These scales are most frequently defined by giving a range of scores, e.g. from 1 to 5 points, or from 0 to 3 points. They are descriptive scales, which use only positive values. In many cases, such an approach might not be sufficient, especially when impacts on the social or natural environment are submitted to analysis. It seems justifiable to use also negative values when negative impacts of development projects are likely to appear. Hence, an original method called the Impact Method, developed by one of the authors of this article, was used in this study. The scale it includes comprises positive and negative values.

Another problem that is not solved by the use of generally known multi-criteria methods is how to assess individual impacts inclusive of their range. In many cases, the assessment depends on the appraiser's

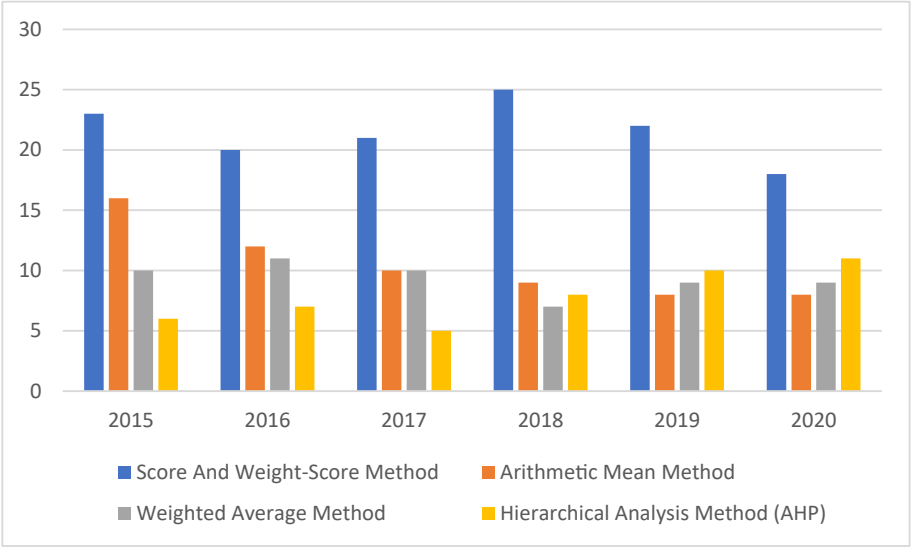


Fig. 3. The most frequently used methods to assess variants of investments implemented in the north-east Poland in 2015–2020. (Source: own studies).

Table 1
An example of a survey prepared for the Indicator Method.

Criterion	Expected impact of the assessed variant on the natural environment										
	−5	−4	−3	−2	−1	0	1	2	3	4	5
A1											
A2											
A3											

discretion, and it does not specify whether a given impact is direct, related to local nuisances, or indirect, pertaining to a wider area. In their publications dealing with this subject, the authors of this article were unable to find a methodology that would include negative values in an assessment scale, additionally acknowledging direct and indirect impacts. Another advantage of the proposed methodology is the grouping of evaluation criteria and ascribing them to the consecutive stages of a building's life cycle.

The review of the literature on the subject [44–46] shows that the multi-criteria analysis methods used to evaluate variants are generally known, and procedures described by many authors [43,54] are used. The innovativeness of the method for the evaluation of variants proposed by the authors lies in:

- making an assessment based on a wider scale, with positive and negative values;
- relating the main criteria to the stages of the facility's life cycle;
- distinguishing between direct and indirect (local and global) impacts in the assessments, which is very important because in many situations impacts perceived as positive from the local perspective may turn out to be negative impacts on a more global sense, and vice versa.

Moreover, the above method, developed by one of the authors of the article, has not been employed in this type of evaluation yet. It also allows to fill the previously described gap in the methodology of environmental impact assessment of construction projects.

4. Research aim, methodology and scope

The aim of the article is to show the possibility of evaluating variants using the methodology that includes positive and negative evaluation values while distinguishing between direct and indirect impacts (local and global ones), and where an appraisal is based on groups of criteria assigned to stages in a building's facility life cycle. In brief, the use of an innovative approach called the Indicator Method was demonstrated.

The proposed method, owing to the inclusion of negative values, provides a wider scale of assessments and thus allows for greater diversity of the final appraisal. In order to show how to apply the proposed method, variants of one structural element of a production hall were assessed.

4.1. Indicator method

The Indicator Method (IM) was developed in 2014–2019 by one of the authors of this research paper, based on her research experience spanning many years, and described in several articles [55,56]. This method allows the user to take into consideration local and global impacts and effects, and the evaluation can also include negative scores accounting for adverse effects of the development in question. Same as in other methods, the starting point for further analysis is defining criteria which the decision maker will use to evaluate all variant solutions of a building development. This stage in the procedure, same as in other cases, necessitates the participation of experts [57]. Their contribution is needed so as to learn about expectations and worries related to the planned investment, and opinions of several experts will provide an insight into the different ways in which the reality and ongoing processes are perceived. The survey addressed to experts differs from other questionnaires because it lets the respondents assign both positive and negative scores. When evaluating variants of a planned development with this method, the criteria included in the assessment procedure are evaluated on a scale from −5 to +5. This approach enables us to evaluate both positive and negative effects of the evaluated development. Table 1 shows how adverse effects of the analysed investment can be evaluated. Obviously, this evaluation takes into account both direct and indirect impacts.

The subsequent stage in an assessment made with the Indicator Method, same as in other approaches, is to determine the importance of the previously described criteria, and the weights define the priorities, which can be met by the analysed variants to different degrees.

The Indicator Method uses matrices (constructed in the form of tables), where individual impacts on the natural environment are described, and consecutive criteria are assigned weights (significance of the impact on nature). The total of all the information we are interested in encompasses data on all the analysed variants. Table 2 explains the principles of constructing a matrix. The number in the upper left-hand corner of each cell describes the direct effect, while the one in the bottom right-hand corner pertains to the indirect effect on a given compartment of the natural environment. Sometimes, these influences are described as local and global ones. Then, direct impacts are identified as local ones and indirect impacts are considered to be global ones. In the centre of the matrix, there is the sum of effects multiplied by the weight.

The weighted partial assessment of the impact on the environment produced by the *i-th* criterion of the *j-th* variant is calculated from the equation (1):

$$Q_{ij} = \left(P_{ij} + R_{ij} \right) W_i \tag{1}$$

where: P_{ij} - direct effect of the subsequent variant in the context of criterion A.

R_{ij} - indirect effect of the subsequent variant in the context of criterion A;
Wi – weight of criterion A.

The sum of particular effects (Q_{ij}) is the partial assessment of the impact of a given variant on the natural environment. Analysis is conducted for each criterion separately.

4.2. Methodology of the research

In order to achieve the research aim, the methodology composed of six steps was implemented:

1. perusal of the literature and source materials, and identification of the research problem,

Table 2
Calculation matrix used in the Indicator Method.

No.	Analysed criterion	Variant 1			Variant 2			Variant 3			Weight of criterion
1	A1	P_{11}	Q_{11}	R_{11}	P_{12}	Q_{12}	R_{12}	P_{13}	Q_{13}	R_{13}	W_1
2	A2	P_{21}	Q_{21}	R_{21}	P_{22}	Q_{22}	R_{22}	P_{23}	Q_{23}	R_{23}	W_2

Table 3
Review of the references concerning roof construction.

No	Authors	Title	Journal/ Publisher	Number/ date
1	Islam, M. S., Chui, Y. H., Altaf, M. S.	Design and Experimental Analysis of Connections for a Panelized Wood Frame Roof [58].	*System. Buildings*	*12*/6, (2022).
2	Islam, M. S., Chui, Y. H., Altaf, M. S.	A holistic design approach for innovative panelized light-wood frame roof construction [59].	*Proceedings of the WCTE.*	(2021).
3	Stark, N.,Cai, Z.	Wood - based composite materials: panel products, glued laminated timber, structural composite lumber, and wood–nonwood composites [60].	*Chapter 11 in FPL-GTR-282, 11-1.*	(2021).
4	Villar-García, J. R., Vidal-López, P., Rodríguez-Robles, D., Guaita, M.	Cost optimisation of glued laminated timber roof structures using genetic algorithms [61].	*Biosystems Engineering,*	187 (2019).
5	García, H., Zubizarreta, M., Cuadrado, J., Osa, J. L	Sustainability improvement in the design of lightweight roofs: A new prototype of hybrid steel and wood purlins [62].	*Sustainability,*	*11*(1), (2018).
6	Gupta, R. S.	Principles of Structural Design: Wood, Steel, and Concrete [63].	*Crc Press.*	(2019).
7	Hassan, O. A., Johansson, C	Glued laminated timber and steel beams: A comparative study of structural design, economic and environmental consequences [64].	*Journal of Engineering, Design and Technology.*	(2018)
8	Lapidus A. A., Makarov A. N.	Model for the potential manufacture of roof structures for residential multi-storey buildings [65]	*Procedia, Engineering*	153 (2016).
9	Breyer, D. E., Cobeen, K. E., Fridley, K. J., & Pollock, D. G.	Design of wood structures—ASD/ LRFD [66].	*McGraw-Hill Education.*	(2015).
10	Franke, S., Franke, B., Harte, A. M.	Failure modes and reinforcement techniques for timber beams–State of the art [67].	*Construction and Building Materials*	*97* (2015).
11	Kloiber, L. A., Eckmann, D. E., Meyer, T. R., Hautzinger, S. J.	Design considerations in cable-stayed roof structures [68].	*Modern Steel Construction,*	*44*(3) (2004).
12	Newman, A.	Metal building systems: design and specifications [69].	*McGraw-Hill Education.*	2004
13	Aoki, H.	Establishment of design standards and current practice for stainless steel structural design in Japan [70].	*Journal of Constructional Steel Research,*	*54*(1) (2000).

2. a survey study, allowing the determination of which factors related to the impact of a building on the environment are significant at the different stages of the evaluation,
3. preparation of a list of criteria, including their division according to the life cycle of a building, and determination of the importance of each criterion,
4. development of variant designs for roofing a production hall,
5. evaluation of a model solution using the Indicator method,
6. analysis of the results and an evaluation of the adopted design concept.

5. An example of the application of the methodology

The construction industry is characterised by an extremely high diversity and uniqueness of solutions. This allows for free shaping of buildings that play various functions, but also for satisfying the expectations of future users of buildings. The development of building structures involves the use of a variety of construction, material and technological solutions. This is conducive to the effective development of variants of planned investments. Different variants of solutions may concern the development of the concept of entire facilities, but also variants of solutions for structural elements, which are also often developed, can be found. For example, the solutions for the construction of external walls, apart from structural aspects, may take into account the insulation performance and the foundation of the building, as well as the economic efficiency aspects of the investment. The roof structure is probably this component of a building that presents the greatest variety of design, technological and material solutions. This applies to residential and service buildings as well as production facilities, such as production halls.

Details regarding the material and construction solutions for roofs of buildings are presented in numerous publications, attesting to the great variety of solutions. Some examples are given in Table 3.

The examples collated in the table confirm the wealth of structural solutions for constructing roofs, which encouraged the authors to choose a roof of a production hall as a model to present the use of the Indicator Method. All elements and the entire structure of a building can be assessed in terms of their impact on the broadly understood environment throughout the entire life cycle of the building, which can be captured with the help of the Indicator Method.

5.1. Characteristics of the developed variants

An analysis was made of different solutions for the roofing of a production hall with the dimensions of 21 × 71 m, and the ridge height of 11.12 m, and an office building with utility rooms. The above are the dimensions of the indoor space, needed for convenient performance of technological processes. Roofs of halls are one of the construction elements of this kind of structures. Nowadays, with the advances in computer-aided design and improved quality of different building materials, more design solutions as well as methods for analysing designed structures are possible. The choice of the type of hall covering depends on many factors, e.g. its span, the height of the space needed for production purposes, whether internal transport is foreseen, if the planned roof will be a purlin one or not, etc.

The literature describes optimisation processes involved in the selection of different material and construction solutions for roof girders and trusses [71–73]. A large number of factors involved means that before one solution is ultimately accepted, the constructor must take into consideration all pertinent factors and their mutual dependences. Lattice or plate girders, made of wood, steel, reinforced concrete or even prestressed concrete, are popular [74–76]. With small spans, up to 9 m, cold-formed steel profiles are a common choice, but for larger roofs (spanning over 12 m), truss systems are a preferred solution.

In this study, 3 variants were developed for roofing a production hall with the dimensions of 21.2 m × 71.2 m and the inner ridge height of 11.21 m. The sections were chosen according to the guidelines and recommendations found in the reviewed references. In each case, the roof slope angle was 4°. Loads were accumulated according to the

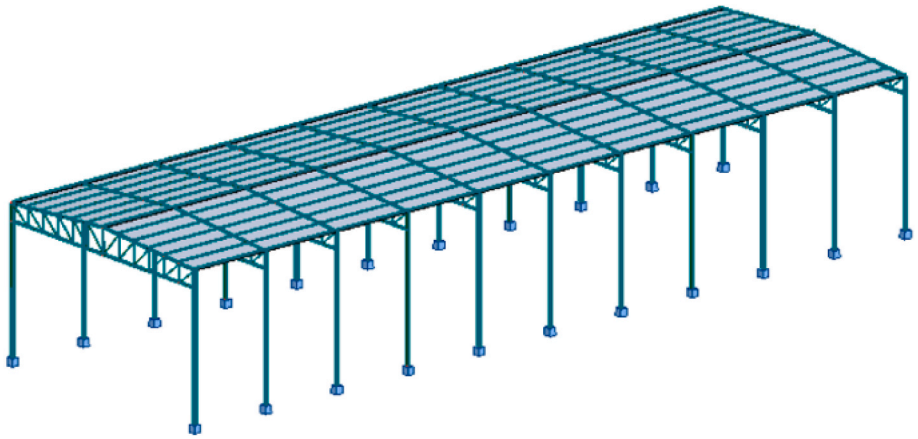


Fig. 4. Static scheme of a steel and solid wood structure (source: Jurczak M.).

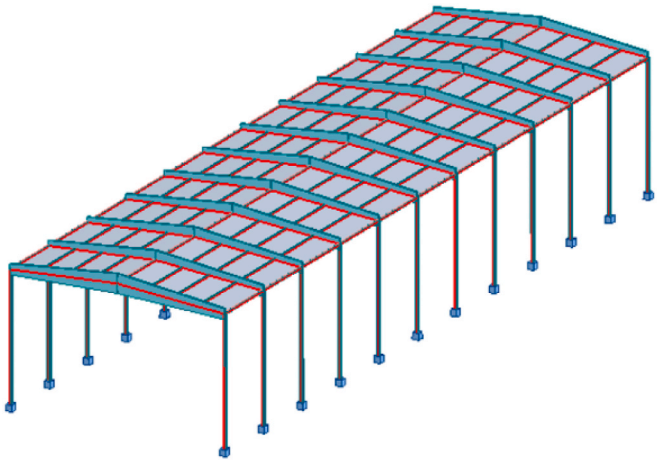


Fig. 5. Static scheme of a structure made of glued laminated timber (source: Jurczak M.).

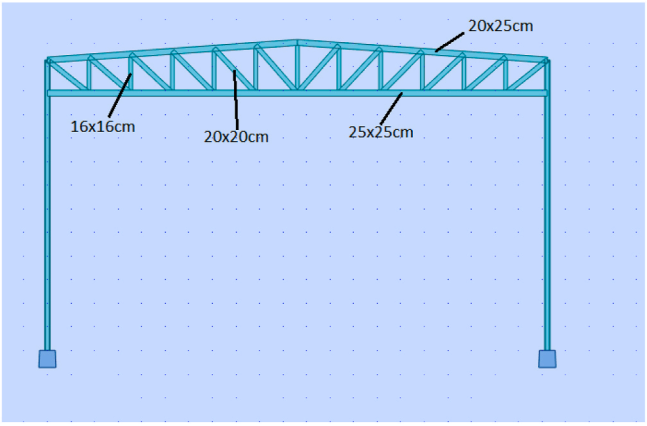


Fig. 6. Wooden truss girder - option 2 (source: Jurczak M.).

standards [77–79] and their combinations were entered into calculations. Cross-sections of elements of the truss were chosen by entering the construction systems into the commercially available software Autodesk Robot, using the results obtained as a decision-making support.

Variant 1. (Fig. 4) – this solution consists of a steel truss with the spacing of 7.12 m, constructed from hot-rolled sections. In the design, the transoms and the lower chord members were made from HEA180 sections, purlins – from HEA120 sections, and posts and cross braces – from steel flat bars RP80x80x4. The N-type lattice was chosen (Fig. 6) with the lattice height ranging from 1.40 to 2.14 m. The calculations took account of the lateral-torsional buckling of the 'beam' type elements, and the flexural-torsional buckling for the posts.

Variant 2. (Fig. 5) The second variant cover had the same shape and dimensions as in variant 1, but a different material was used, namely class C27 solid wood. The dimensions of the elements were as follows: purlins 14 × 20cm, transoms – 20 × 25cm, lower chord – 25 × 25cm, posts – 16 × 16cm, and cross braces – 20 × 20cm.

Variant 3. In the third case, full-walled transoms were chosen, composed of a double-tapered beam from class GL32h glulam (laminated timber). The section with the height in the range of 100–180 cm and width of 20 cm was used, purlins consisted of rectangular beams size 14 × 20cm (Fig. 6). The girder beams were spaced at 6.47 m.

5.2. Determination of the list of assessment criteria and description of the study

In order to elaborate a list of assessment criteria, the life cycle of the building for which the roof design variants were developed was analysed. The first step was to identify factors important from the viewpoint of the impact on the environment due to the extraction and processing of raw materials and building materials. An important element of this stage is the option to take into account recycled materials. The next important step in the evaluation of variant designs is whether the construction works will be carried out in a way that will reduce harmful emissions, noise or transport. Another question is whether the duration of this impact can be shortened.

The longest period in a building's life is its operation and maintenance phase. For production halls, it is necessary to take into account the specific conditions arising from the type of production carried out inside these buildings, but equally important are the measures taken to maintain the construction in a proper technical condition, the ease and frequency of maintenance works, etc. Finally, it is necessary to consider the processes connected with the termination of the operation of a building, its end-of-life stage and possible re-use of demolition materials. Following a review of the literature, interviews with experts, and the evaluation of design variants, a list of factors which can serves as the basis for the elaboration of assessment criteria was drawn up.

To determine the list of criteria, a study in the form of an in-depth interview addressed to a group of 56 experts was made. The experts were representatives of construction designers, investors and users of similar buildings. This stage of the study led to the determination of 18 factors which appeared at all stages in the life cycle of the building. The next step leading to the assessment of the importance of the criteria consisted of a survey addressed to the same experts. This study was

Table 4
Assessment of the importance of analysed criteria.

No	Control question	Is this aspect important?					Score	Weights
		Very impor-tant (5p)	Impor-tant (4p)	Moderately impor-tant 3p)	Not very important (2p)	Unim-portant (1p)		
1	Extraction of raw materials	x					5	0.0820
2	Process of manufacturing and processing		X				4	0.0656
3	Energy consumed for production	x					5	0.0820
4	Use of recycled materials			X			3	0.0492
5	External transport			X			3	0.0492
6	Transport on construction site				x		2	0.0328
7	Assembly process on construction site			X			3	0.0492
8	Organisation of processes and construction site				x		2	0.0328
9	Resistance to external environment factors			X			3	0.0492
10	Resistance to moisture from production processes			X			3	0.0492
11	Resistance to fire			X			3	0.0492
12	Durability (trouble-free operation time)		X				4	0.0656
13	Need (ease and frequency) for maintenance works				x		2	0.0328
14	Emissions to the environment (material preserving chemicals)			X			3	0.0492
15	Ease of disassembly			X			3	0.0492
16	Possibility of re-using		X				4	0.0656
17	Possibility of recycling	x					5	0.0820
18	Harmful waste that must be disposed of		X				4	0.0656
	Sum						61	1

conducted in 2020–2021. The survey allowed the respondents to provide an answer leading to the assignment of weights to the criteria. Table 4 contains the results of this stage of the research.

In the evaluation of the mentioned factors, it is difficult to conclude conclusively which factors are perceived as direct or indirect (local or global) ones. In principle, each of these activities generates some effects on a smaller as well as a broader scale. In many cases, we can notice that local effects can be positive locally but negative on a broader scale, and vice versa. The factors described and taken into account in our study were mostly evaluated as important or very important. Only three factors, transport, construction processes and need for maintenance works, were thought to be less important. The reason could be the traditional approach to an evaluation of the impact on the environment shared by most of the experts. While the issues associated with the production and extraction of raw materials are becoming more widely acknowledged, the problems generated by transport and maintenance works are less often considered by those involved in the construction processes. This study attests to the need for raising the awareness of analyses of the impact of the environment made in the context of a building's life cycle, including also less recognised problems.

This stage of our study led to the determination of the following assessment criteria:

I. Preparation stage (A1)
  1.1. Extraction of raw materials (A11)
  1.2. Process of production and processing (A12)
  1.3. Energy for production (A13)
  1.4. Use of recycled materials (A14)
  1.5. Transport (A15)
II. Construction stage (A2)
  2.1. Transport on the construction site (A21)
  2.2. Process of assembly on the construction site (A22)
  2.3. Organisation of processes and the construction site (A23)
III. Operation (A3)
  3.1. Resistance to the external environment factors (A31)
  3.2. Resistance to moisture from production (A32)
  3.3. Resistance to fire (A33)
  3.4. Durability (time of trouble-free operation) (A34)
  3.5. Need (ease and frequency) of maintenance (A35)
  3.6. Emissions to the environment (material preserving chemicals) (A36)
IV. Demolition (A4)
  4.1. Ease of disassembly (A41)
  4.2. Possibilities of re-use (A42)
  4.3. Possibilities of recycling (A43)
  4.4. Harmful waste that requires proper disposal (A44)

The evaluation of the impact of the mentioned factors on the environment conducted on a local scale as a direct effect may generate a positive result, while on a global scale, as an indirect effect, it could be negative. An example is fire resistance. On a local scale, this is obviously a positive factor, but indirectly the production of chemicals that improve fire resistance may have a negative effect. Contrarily, using recycled materials may have negative effects locally, due to either the aesthetic or strength properties being inferior, while indirectly, on a global scale, this option is the most beneficial one. All the evaluation criteria underwent a similar analysis, and the results are shown in Table 5. The evaluation of the design variants was made on the basis of the Impact Method formulas, presented in section 4.1.

5.3. Discussion of the results

The evaluation made with the Indicator Method, proposed by the authors of this paper, is characterised by the presence of negative values. The impacts on the environment evaluated with this method were predominantly negative, which resulted in values less than zero in both partial assessments and the final assessment outcome. This study shows that the use of steel constructions is most noxious to the environment, which is due to the processes involved in the preparation of this material. Both the extraction and processing of metal ores in smelters require high temperatures, which is evaluated negatively (both negative scores). The other solutions did not score so low. Both the solid wood and glulam constructions were scored the lowest in terms of the acquisition of the raw material (felling of trees), but they were evaluated as less harmful. The few positive scores were related to the possibilities of recycling and re-use of the materials, and to the durability of constructions. The assessment outcome indicates that the solid wood construction could be the best solution from the viewpoint of its impact on the environment.

The visual presentation of the results may help to interpret the

Table 5
Evaluation of the described variants of a construction design, made with the Indicator Method.

No.	Analysed criterion	Variant no 1 (steel structure)			Variant no 2 (solid wood)			Variant no 3 (laminated wood)			Weight
		P	Q	R	P	Q	R	P	Q	R	W
1	A11	-4	**-0.73770**	-5	-2	**-0.40984**	-3	-2	**-0.40984**	-3	0.0820
2	A12	-5	**-0.65574**	-5	-2	-0.19672	-1	-2	-0.32787	-3	0.0656
3	A13	-5	**-0.81967**	-5	0	-0.08197	-1	-2	-0.2459	-1	0.0820
4	A14	3	0.39344	5	-2	-0.09836	0	2	0.14754	1	0.0492
5	A15	-3	-0.29508	-3	-2	-0.19672	-2	-2	-0.19672	-2	0.0492
6	A21	-2	-0.06557	0	-1	-0.03279	0	-1	-0.03279	0	0.0328
7	A22	-3	-0.24590	-2	-1	-0.14754	-2	-2	-0.19672	-2	0.0492
8	A23	2	0.13115	2	3	0.19672	3	3	0.19672	3	0.0328
9	A31	-3	-0.34426	-4	-2	-0.24590	-3	-1	-0.14754	-2	0.0492
10	A32	-5	-0.34426	-2	-3	-0.24590	-2	-2	-0.19672	-2	0.0492
11	A33	-3	-0.39344	-5	-3	-0.24590	-2	-2	-0.14754	-1	0.0492
12	A34	5	0.65574	5	3	0.39344	3	4	0.52459	4	0.0656
13	A35	-3	-0.09836	0	-2	-0.06557	0	-1	-0.03279	0	0.0328
14	A36	-5	-0.39344	-3	-4	**-0.29508**	-2	-2	-0.19672	-2	0.0492
15	A41	2	0.24590	3	3	0.34426	4	1	0.04918	0	0.0492
16	A42	-2	-0.32787	-3	5	0.65574	5	4	0.52459	4	0.0656
17	A43	5	0.81967	5	2	0.49180	4	2	0.24590	1	0.0820
18	A44	4	0.39344	2	5	0.65574	5	3	0.39344	3	0.0656
	Sum **Q**		-2.08197			**0.47541**			-0.04918		1

analysis. Let us present it for variant 1 (v1) – evaluated the lowest, and variant 2 (v2) – which emerged as the best. The vertical axis shows values resulting from the previous procedure assigned to P, Q and R. The vertical axis displays values for 1-P, 2-Q, 3-R. Fig. 7 clearly shows that criterion A14 in the assessment of variant 1 was the only factor in group 1 which scored positively. This criterion stands for the possibility of using recycled materials, which is valid for steel construction as nearly 100% of steel used can be recovered after demolition and re-used after melting. However, the other criteria are rated much worse (−4 and −6), leading to a very low rating of this stage in terms of its impact on the environment. As for v2 – which was the highest rated variant – all scores in this group assume negative values, but much lower than in the case of

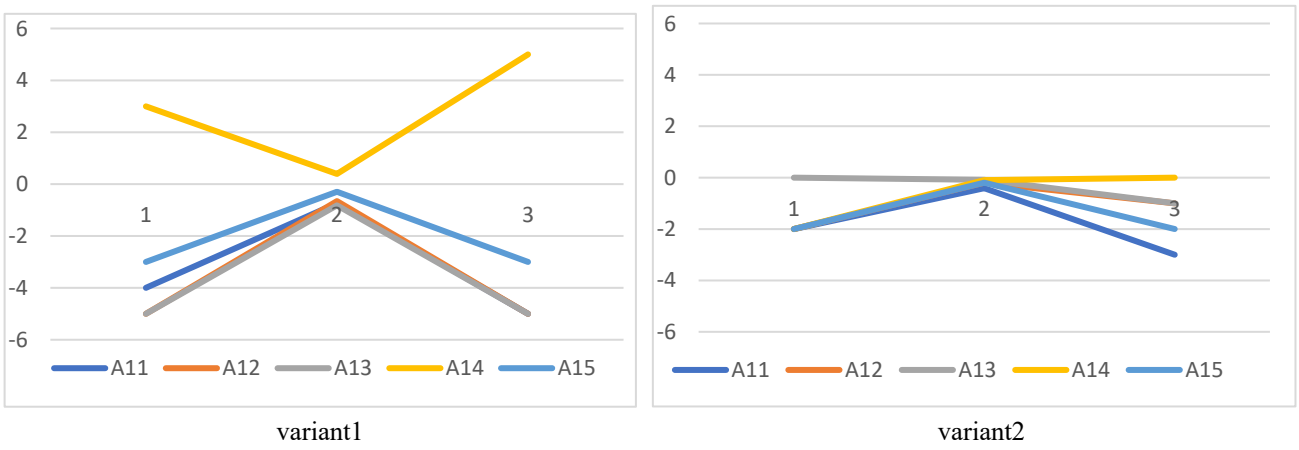


Fig. 7. Visual presentation of results for factors in group 1.

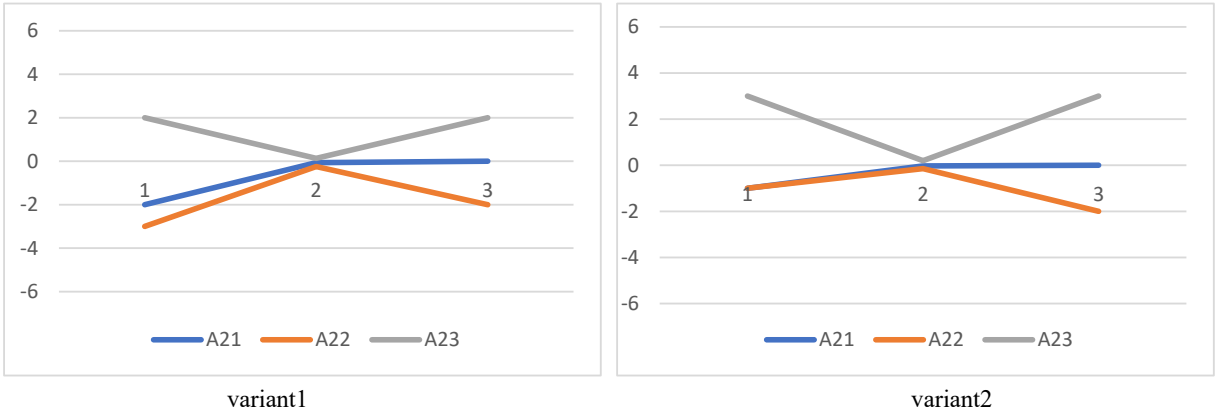


Fig. 8. Visual presentation of results for factors in group 2.

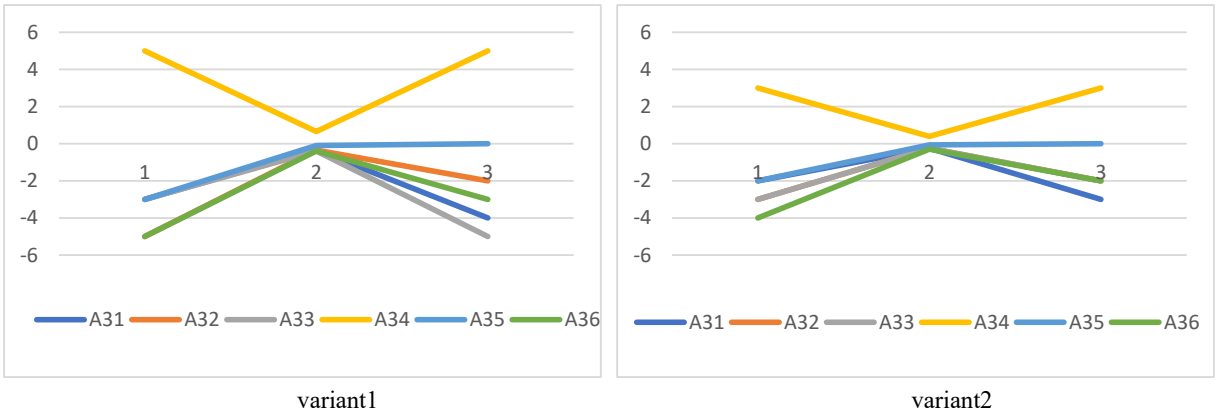


Fig. 9. Visual presentation of results for factors in group 3.

v1. The lowest score is −3. In consequence, the final rating is higher.

In group 2 of the criteria, criterion A23 is evaluated positively for both variants, 1 and 2 (Fig. 8). This criterion applies to the organisation of construction works on a construction site. This is an important factor, which can alleviate the impact on the environment, both locally and globally. The other two criteria were evaluated negatively, slightly worse for v1.

In group 3 of the criteria, one of the lowest scores obtained in the assessment of variants 1 and 2 can be noted (Fig. 9). This was the assessment of criterion A34 – durability of the construction. Steel constructions if maintained correctly can last flawlessly for as long as 100 years. The durability of a construction made of properly protected timber is rated high. The other factors scored negatively, and – same as criteria from group 1 – values assigned to v1 are much lower than the ones earned by v2.

The last group of analysed factors comprises three positive scores for v1 and four for v2. Criterion A43 is the one scoring the highest for v1, and A44 – for v2. Criterion A43 represents the possibility of recycling, and A44 – harmful waste requiring proper disposal (Fig. 10).

For a complete comparison of the variants, the visual interpretation

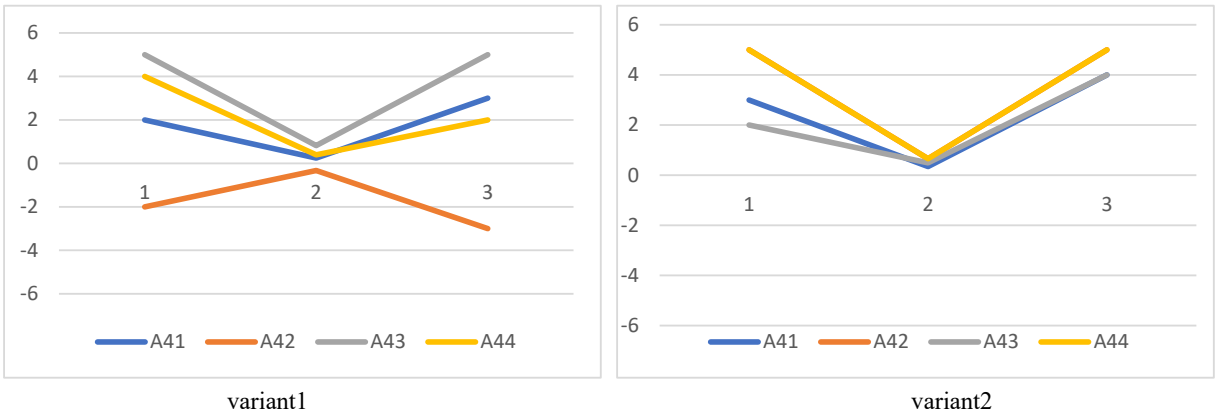


Fig. 10. Visual presentation of results for factors in group 4.

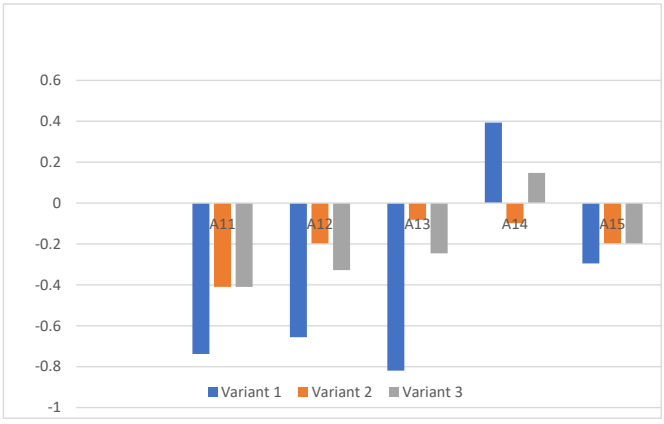


Fig. 11. Visual interpretation of the final assessment of 1 group criteria.

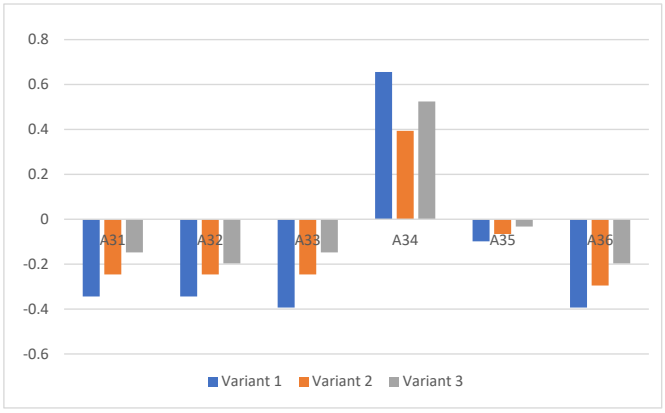


Fig. 13. Visual interpretation of the final assessment of 3 group criteria.

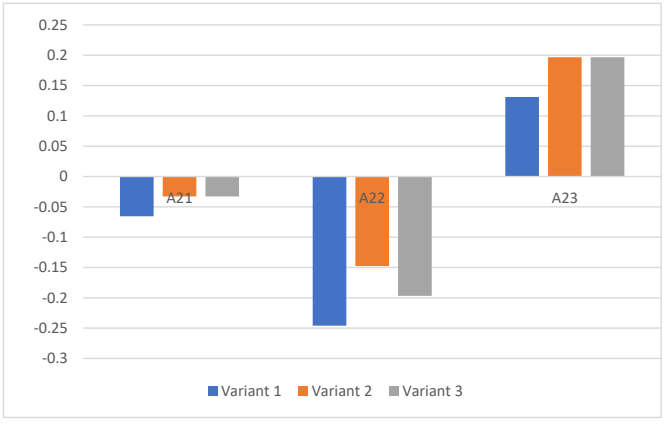


Fig. 12. Visual interpretation of the final assessment of 2 group criteria.

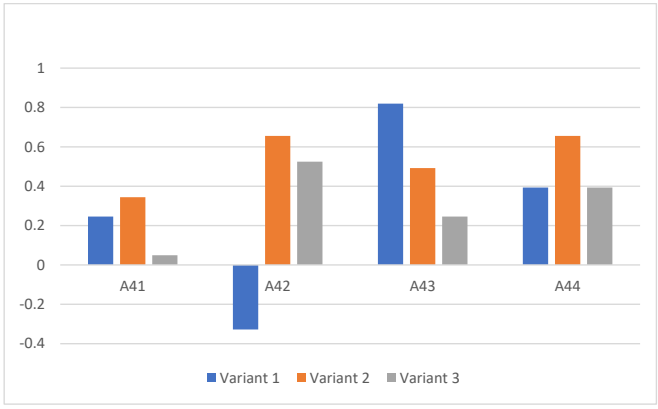


Fig. 14. Visual interpretation of the final assessment of 4 group criteria.

of the final assessment of group 1 criteria is given (Fig. 11).

Negative scores prevail in the assessment of the criteria made for all the variants, but the values assigned to variant 1 were distinctly more negative. Only criterion A14 scored positively, but this did not compensate for the remaining very low scores. Variant 2 obtained the least negative scores in the assessment of criteria A12 and A13.

The evaluation of group 2 of the criteria confirms the lowest scores of variant 2 and the best ones for variant 3 (Fig. 12).

Group 3 criteria dealing with the utilisation of the construction achieved the best values for variant 3 (Fig. 13).

The fourth group of criteria pertaining to the last stage in the life cycle, that is the demolition and disposal of waste, yielded highly discrepant results (Fig. 14). Variant 1 was evaluated well in terms of criterion A43, and obtained only one negative value for criterion A42. Variant 2 achieved very high scores for all the criteria. Variant 3 scored positively for all the criteria, but these values were not very high.

The final outcome of the evaluation (Table 5) indicates that the second variant, that is the solid wood construction, is the best, and the method proposed in this study allows the user to assess impacts of the construction both as negative and positive ones.

The criteria were divided into groups corresponding to the life cycle of a building (Preparation stage Σ (A11÷A15); Construction stage Σ

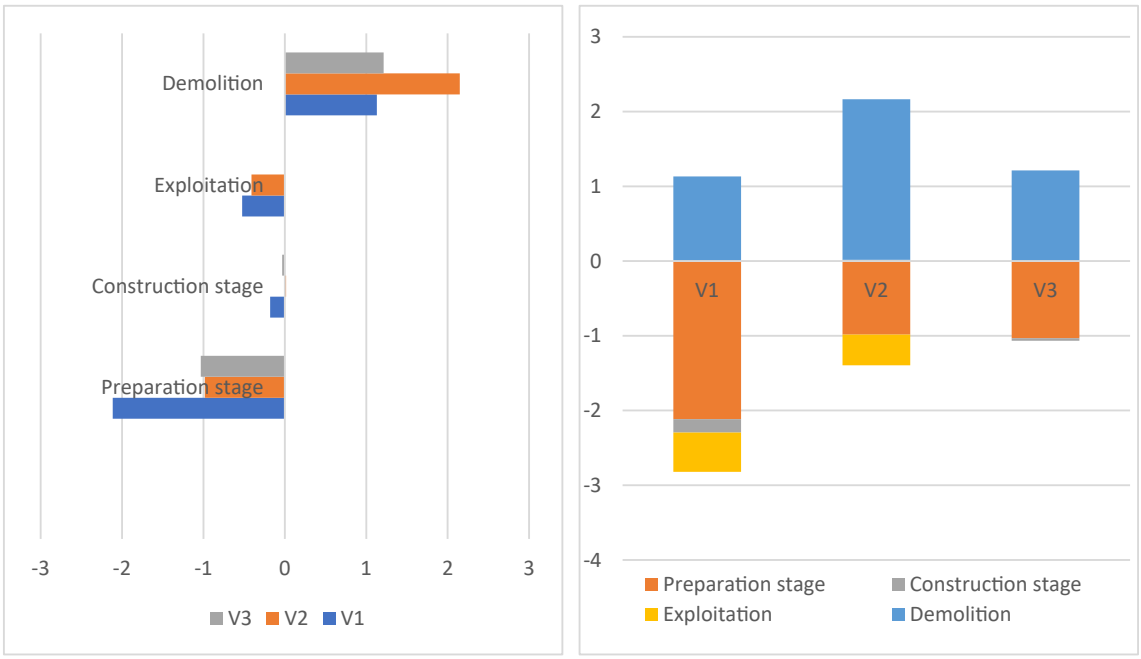


Fig. 15. Assessment of impacts in the subsequent stages of the building's life cycle - graphic illustration.

(A21÷A23); Operation Σ (A31÷A36); Demolition Σ (A41÷A44)). This approach allows one to analyse effects at every step of a life cycle, and to determine which step is the most difficult one and generates the most severe negative impacts. A deeper analysis of the results shows that the stage that is most noxious to the environment is the manufacture of building materials and products, and the negative value in this aspect was assigned to all the three construction variants. This is confirmed by the diagram in Fig. 15.

The results show that the final assessment can achieve a positive or a negative value. This leads to greater differentiation of the assessment results and in the examined case directly indicates the only positively evaluated solution. Presenting the results of an appraisal as shown above allows one to indicate the most troublesome stage and propose solutions that should improve the overall assessment.

The method proposed for the analysis of the building-environment relationship is relatively simple, which facilitates its application in practice. An additional advantage is also the simple interpretation of the results, the ease of identifying the best solution and the assessment of life cycle stages. The limitation of the method is the problem with its application when the assessment criteria cannot be evaluated by taking into account negative and positive impacts. It is also difficult in some cases to determine whether the impact considered in the assessment is local or global (direct or indirect). In such situations, it is necessary to expand the panel of experts by adding more respondents.

The Indicator Method gives many possibilities to carry out assessments of building developments, but it can still be further developed and improved. The authors plan to continue research in which they will focus on refining expert assessments, interpretation of results and extending the application possibilities by emphasising the universality of the method.

6. Summary and conclusions

The case study presented in the article, related to the use of the methodology developed by the authors, is aimed at showing some innovative aspects of the assessment of construction investment variants in terms of their environmental impact.

One of the assumptions was to take into account a wider impact assessment scale inclusive of negative values. This allowed for a precise and effective assessment, and the final results of the appraisal showed greater differentiation than assessments used with traditional methods [43–54]. This outcome provides the basis for more effective decision making. The research showed that an assessment scale with negative values was well received by the experts participating in the survey. In most impact assessments, a negative score was assigned, which confirms the usefulness of the proposed research methodology.

Another innovative aspect of the assessment presented above is the simultaneous consideration of direct and indirect impacts. It is important from the point of view of the evaluator and allows to specify what the next value included in the questionnaires concerns. Specifying these two areas of assessment allows for its more precise presentation, for example showing that a positive reception of an object in the local area may be accompanied by wider negative impacts, and vice versa. Such an approach cannot be found in the literature, indicating it constitutes a novel approach to assessment.

Another important aspect is that the main criteria are referred to the stages of the building's life cycle. This approach makes it possible to analyse the impacts at each stage and to determine which of them is the most difficult and generates the most severe negative impacts. The method applied for the evaluation was not used in this type of evaluation, developed by one of the authors of the article. The model evaluation presented in the article attests to the usefulness of the Indicator Method, and the analysis of the results obtained proves that it can fill the knowledge gap in the methodology of environmental impact assessment of construction projects.

The conducted study confirms that the proprietary Indicator Method, which takes into account both positive and negative values, the assessment of direct and indirect impacts, and the life cycle stages of a building in relation to the impact assessment, is capable of indicate problematic areas in an assessment, and the assessment scale inclusive of negative scores allows for more precise assessments.

CRediT authorship contribution statement

Elżbieta Szafranko: Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Marlena Jurczak:** Writing – original draft, Visualization, Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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